

UNIFORM VERTEX-PUSH CHAINS ON GRAPH ORIENTATIONS: SPECTRUM, PERIOD, AND COMPONENT-SIZE INVERSION

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ABSTRACT. Let a step on an oriented finite simple graph choose a vertex uniformly and reverse every incident edge. Restricted to one push orbit, the chain is an abelian quotient walk. We give its complete relation presentation and orbit size, then compute every transition eigenvalue and multiplicity through a component sign-count polynomial. Character orthogonality yields the return probability at every time. The chain has period two exactly when every connected component has even order; all other cases are aperiodic. The spectrum also has a precise inverse consequence. Given the ambient vertex number, it determines the multiset of component orders. The proof factors the spectral polynomial into even-binomial polynomials and orders their nearest negative roots; a collision argument makes the multiplicity peel exact, and the known total recovers isolated vertices. Internal component adjacency and the starting orientation remain unidentifiable. The vertex-push operation, finite-abelian Fourier diagonalisation, and generic uniform stationarity are treated as established inputs rather than contributions.

1. SCOPE, CHAIN, AND RESULT

Reversing every arc incident with a chosen vertex is the established vertex-push operation; its orientation setting and push-equivalence language predate the present chain analysis [1, 2]. Fourier analysis on finite groups is likewise standard [3]. We assign those inputs, and generic uniform stationarity for symmetric finite walks, zero contribution credit. The purpose here is the narrow exact package in theorem 1.1; no novelty or priority claim is made.

Let $G = (V, E)$ be a finite simple graph with $n = |V| \geq 1$. Its connected components are $C_1, \dots, C_c$, with orders $s_i = |C_i|$. Fix an orientation ω_0 . A *push* at v reverses every edge incident with v . Starting anywhere in the push orbit of ω_0 , choose $v \in V$ uniformly and push it at each discrete time. Write X_t for the resulting chain.

Define

$$(1) \quad B_s(x) = \sum_{\substack{0 \leq j \leq s \\ j \text{ even}}} \binom{s}{j} x^j, \quad M_G(x) = \prod_{i=1}^c B_{s_i}(x).$$

Theorem 1.1. *For the uniform vertex-push chain on one orbit:*

- (1) *pushes commute, the product of all pushes in each component is the identity, and these are the only relations. Every orbit has 2^{n-c} orientations. The chain is irreducible there and its stationary law is uniform;*
- (2) *for $0 \leq k \leq n$, the transition eigenvalue*

$$(2) \quad \lambda_k = \frac{n - 2k}{n}$$

has multiplicity $[x^k]M_G(x)$. Thus, for every integer $t \geq 0$,

$$(3) \quad \mathbb{P}(X_t = X_0) = 2^{-(n-c)} \sum_{k=0}^n [x^k]M_G(x) \left(\frac{n - 2k}{n} \right)^t;$$

- (3) *the period is two if every s_i is even, and is one otherwise;*
- (4) *given n , the transition spectrum, including multiplicities, determines the multiset $\{s_1, \dots, s_c\}$. It does not determine internal component adjacency or the starting orientation.*

The four assertions are proved in order below.

2. THE PUSH ORBIT

Use $\mathbb{F}_2 = \{0, 1\}$ and encode an orientation relative to ω_0 by the set of edges whose directions were reversed. Define the mod-two cut map

$$(4) \quad \delta : \mathbb{F}_2^V \longrightarrow \mathbb{F}_2^E, \quad (\delta a)_{uv} = a_u + a_v \quad (uv \in E).$$

Lemma 2.1. *The kernel of δ is*

$$(5) \quad \ker \delta = \text{span}_{\mathbb{F}_2} \{\mathbf{1}_{C_1}, \dots, \mathbf{1}_{C_c}\}.$$

Proof. The equality $\delta a = 0$ says $a_u = a_v$ on every edge uv . Thus a is constant along paths and hence on each connected component. Conversely, a vector constant on every component plainly has $a_u + a_v = 0$ on every edge. The component indicators have disjoint nonempty supports, so they are independent and span exactly these componentwise constant vectors. $\square$

Proposition 2.2. *The push orbit is a torsor for*

$$(6) \quad A_G = \mathbb{F}_2^V / \text{span}\{\mathbf{1}_{C_1}, \dots, \mathbf{1}_{C_c}\}, \quad |A_G| = 2^{n-c}.$$

On this torsor the transition is addition by a uniformly chosen class of a coordinate vector e_v . In particular, the chain is irreducible and its unique stationary law is uniform.

Proof. Pushing v adds δe_v to the orientation vector. Such additions are commuting involutions. Pushing a subset $S \subseteq V$ is the identity exactly when $\mathbf{1}_S \in \ker \delta$. By lemma 2.1, these subsets are precisely unions of whole components. This proves both the stated component relations and the absence of any others. The induced action of A_G is therefore free and transitive on the orbit, and (6) gives its size.

The coordinate classes generate A_G , so the walk is irreducible. Each transition $a \mapsto a + [e_v]$ is an involution; consequently the transition matrix is symmetric and doubly stochastic. Uniform measure is stationary, and irreducibility makes it unique. $\square$

3. COMPLETE SPECTRUM AND RETURN LAW

We give the short specialization of finite-group character diagonalisation because it fixes every multiplicity. A character of A_G is specified by signs $\varepsilon_v \in \{1, -1\}$ through

$$(7) \quad \chi_\varepsilon([a]) = \prod_{v: a_v=1} \varepsilon_v.$$

It descends from $\mathbb{F}_2^V$ exactly when

$$(8) \quad \prod_{v \in C_i} \varepsilon_v = 1 \quad (1 \leq i \leq c).$$

Theorem 3.1. *The spectrum and return probabilities are given by (2)–(3), including multiplicities.*

Proof. Condition (8) is exactly triviality on every relation in (6); hence the legal sign assignments give all $|A_G|$ characters. For the transition operator P and any such character,

$$(9) \quad P\chi_\varepsilon(a) = \frac{1}{n} \sum_{v \in V} \chi_\varepsilon(a + [e_v]) = \left(\frac{1}{n} \sum_{v \in V} \varepsilon_v \right) \chi_\varepsilon(a).$$

If k signs are negative, the factor in parentheses is $(n - 2k)/n$. Condition (8) says that the number of negative signs in each component is even. Component C_i therefore contributes the factor $B_{s_i}(x)$ in (1); multiplying the independent choices shows that exactly $[x^k]M_G(x)$ characters have weight k . Characters form a basis, so this is the complete spectrum.

Finally, character orthogonality on A_G gives the identity coefficient of the t -step convolution kernel as

$$\frac{1}{|A_G|} \sum_{\chi} \hat{P}(\chi)^t.$$

Substituting $|A_G| = 2^{n-c}$ and grouping the characters by k proves (3). Translation invariance makes this return probability independent of the initial point in the orbit. $\square$

4. THE EXACT PERIOD DICHOTOMY

Proposition 4.1. *The chain has period two if every component order is even, and period one if some component order is odd.*

Proof. Suppose first that every s_i is even. Coordinate parity

$$[a] \mapsto \sum_{v \in V} a_v \pmod{2}$$

is well defined on A_G : adding any relation $\mathbf{1}_{C_i}$ changes the sum by the even number s_i . Every push adds one coordinate vector and toggles this parity, so every return time is even. Pushing any chosen vertex twice returns in two steps. The period is therefore two.

Now let C_i have odd order s_i . Pushing every vertex of C_i once, in any order, is the component relation and gives a closed walk of odd length s_i . There is also a two-step closed walk obtained by pushing one vertex twice. Both prescribed sequences have positive probability. Their lengths have greatest common divisor one, so irreducibility implies period one. For $s_i = 1$ the first closed walk is simply the self-loop produced by pushing an isolated vertex. $\square$

5. INVERTING THE SPECTRUM

Because n is known, the values $(n - 2k)/n$ are distinct as k varies. Thus the spectrum and its multiplicities recover every coefficient of M_G . Since M_G is even, write

$$(10) \quad E_s(y) = \sum_{r \geq 0} \binom{s}{2r} y^r, \quad Q_G(y) = \prod_{i=1}^c E_{s_i}(y), \quad M_G(x) = Q_G(x^2).$$

Lemma 5.1 (Nearest-root separation). *For every integer $s \geq 2$, all roots of E_s are simple and negative. The root nearest zero is*

$$(11) \quad \rho_s = -\tan^2\left(\frac{\pi}{2s}\right).$$

If $2 \leq r < s$, then $|\rho_s| < |\rho_r|$ and $E_r(\rho_s) \neq 0$.

Proof. For $t \in \mathbb{R}$ the even-binomial identity gives

$$(12) \quad E_s(-t^2) = \frac{(1+it)^s + (1-it)^s}{2}$$

$$(13) \quad = (1+t^2)^{s/2} \cos(s \arctan t).$$

For $t > 0$, its zeros occur at

$$(14) \quad t = \tan\left(\frac{(2j+1)\pi}{2s}\right), \quad j \geq 0, \quad 2j+1 < s.$$

There are $\lfloor s/2 \rfloor = \deg E_s$ such values. They are positive and distinct, so their negative squares are distinct roots and exhaust E_s . In particular all roots are simple, and $j = 0$ gives (11).

If $r < s$, then $0 < \pi/(2s) < \pi/(2r) < \pi/2$. Strict increase of the tangent on this interval proves $|\rho_s| < |\rho_r|$.

It remains to exclude a root collision, rather than infer it merely from this ordering. Put $t_s = \tan(\pi/(2s))$. If $E_r(\rho_s) = 0$, then (12) for r yields an integer $j \geq 0$ such that

$$r \arctan t_s = \frac{(2j+1)\pi}{2}.$$

Here $\arctan t_s = \pi/(2s)$, because that angle lies in $(0, \pi/2)$. Consequently $r/s = 2j+1$. This is impossible: $r/s < 1$, whereas $2j+1 \geq 1$. Hence no smaller-order factor vanishes at ρ_s . $\square$

Theorem 5.2 (Component-order inverse). *Given n , the transition spectrum determines the multiset of connected- component orders.*

Proof. As observed above, the known- n spectrum first recovers Q_G . If this polynomial is nonconstant, let s be the largest component order at least two. By lemma 5.1, ρ_s is closer to zero than the nearest root of every smaller-order factor. It is therefore the root of Q_G nearest zero. No factor of smaller order shares it, by the final part of that lemma, and the root is simple within each copy of E_s . Its multiplicity in Q_G is thus exactly the number of components of order s . The strict ordering of $|\rho_s|$ identifies s itself.

Divide out all of those copies of E_s and repeat. This finite procedure recovers every component order at least two, with multiplicity. The remaining factor is constant because $E_1(y) = 1$. Isolated vertices are therefore recovered from the known ambient order as

$$(15) \quad \#\{i : s_i = 1\} = n - \sum_{i: s_i \geq 2} s_i.$$

If $Q_G = 1$ from the outset, (15) gives n isolated vertices. $\square$

6. INVERSE BOUNDARY AND EXACT CONTROLS

Proposition 6.1. *The transition spectrum does not determine internal component adjacency or the starting orientation.*

Proof. Formula (1) depends on G only through its component orders. For example, the path P_4 and complete graph K_4 are nonisomorphic connected graphs but have the same polynomial $B_4(x)$ and hence the same spectrum. Changing the starting orientation translates the cut image to another affine push orbit; the quotient presentation and transition eigenvalues are unchanged. Thus neither datum can be reconstructed from this spectrum. $\square$

The deterministic verifier accompanying this note uses integers and exact rational arithmetic only. Its principal finite controls are summarized in table 1.

TABLE 1. Exact counterexample pressure; finite enumeration is not a proof.

control	coverage
labelled simple graphs, $1 \leq n \leq 5$	1,099
push-orbit states visited	14,149
return-recurrence state cells, $0 \leq t \leq 6$	71,874
component partitions, fixed totals $1 \leq n \leq 30$	28,628
exact compressed-factor divisions	144,024
total exact assertions	473,328

These checks cover the cut kernel and orbit size, sign multiplicities, direct return recurrences, period boundary, fixed-total spectral signatures, factor peeling, isolate residual, and two nonidentifiability witnesses. They use no floating point or sampling and make no novelty or owner-clearance claim. No figure is included: the quotient, factor, and root relations are exact one-line formulas, while a drawing or numerical root plot would add no evidence. External status remains `HOLD_EXTERNAL`.

REFERENCES

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